Correcting a learned physical invariant improves world-model rollouts

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Abstract

Probing asks whether a model represents a quantity by fitting a readout to it. Its known weakness is that accuracy mixes what the representation stores with what the probe can learn, and the accepted remedy is better probe hygiene. We show the remedy addresses the wrong failure. In a DreamerV3 world model trained only on pendulum video, a polynomial readout fitted to *true* physical energy reaches a correlation of 0.9999 — yet the direction it identifies is one the model's own transition does not preserve, and enforcing it during imagination makes the model's physics worse. A label-free search that tracks energy slightly less well, but is 6.7× better preserved by the transition, improves it by 42%. We then explain the gap. The training data is generated by a symplectic integrator, which conserves not the textbook Hamiltonian H but a shadow $\tilde{H} = H + \mathcal{O}(\Delta t)$. Sweeping the regression target over a one-parameter family containing both signs and wrong magnitudes, the direction best conserved by the learned transition sits at exactly the coefficient the integrator predicts, with no free parameters, on 3 of 3 models and 12× better than the wrong-sign control — and correcting that $\mathcal{O}(\Delta t)$ term flips the intervention from harmful to helpful on every model. The perfect probe was fitted to a quantity the generating process does not conserve, which no probe hygiene detects, because nothing is wrong with the probe. Whether a latent encodes a quantity and whether the dynamics preserve it are different properties, and they dissociate on five further axes — untrained models, out-of-distribution states, a second architecture, an untrained transition, and actuation, where a model that demonstrably uses its actions carries an energy-like scalar whose motion the action's power explains 0.31% of — always in the same direction, with decodability surviving where dynamical structure fails. A statistic on the transition operator separates them in every case, and the quantity it selects is causally deployed: enforcing it cuts drift in decoded physical energy by 55–76% on unseen trajectories, with 0 of 60 magnitude-matched random directions beating it; setting it steers the imagined world's true energy (rank correlation 0.80–0.91); and the effect survives an interchange 50 steps into autonomous imagination. It is also bounded — to the training distribution, and to RSSM-like architectures.

1 Introduction

Probing is how we ask whether a model represents something. Fit a readout from the representation to the property of interest; if it succeeds, the property is there. The method's known weakness is that probe accuracy mixes what the representation stores with what the probe can learn, and the accepted remedy is better probe hygiene — control tasks, selectivity, randomly initialised baselines [Hewitt and Liang, 2019, Belinkov, 2022].

We show that the remedy addresses the wrong failure. On a DreamerV3 world model trained only on pendulum video, we fit a polynomial readout to *true* physical energy. It reaches a correlation of 0.9999: as a probe it could not do better. Yet the direction it identifies is one the model's own transition does not preserve, and enforcing it during imagination makes the model's physics *worse*. A label-free search that tracks energy slightly less well, but is 6.3× better preserved by the transition, improves it by 42%.

The failure is not in the probe. It is that whether a latent *encodes* a quantity and whether the model's dynamics *preserve* it are different properties of the same representation, and no amount of probe methodology separates them — because the probe is already perfect.

Measuring the operator instead. The discriminating measurement is not on the representation but on the transition: apply one autonomous step at a real encoded state and ask how much a candidate scalar moves (Equation 4). This single statistic separates trained from untrained models ($74\times$), conservative from dissipative ($39\times$, no overlap), in-distribution from out (55–$267\times$), one architecture from another ($767\times$), and a label-free scalar from a supervised probe of the same quantity ($6.3\times$). On every one of those contrasts a probe reports success.

The dissociation is directional in each case we can construct: decodability is retained where conservation fails. Probing therefore over-reports, and it over-reports in the situations that matter — outside the training distribution, in architectures that predict well without conserving, and when the quantity is supplied by supervision rather than found in the dynamics.

The quantity the operator selects is used by the model. A conserved-looking scalar could be an artifact of the search. Three interventions say otherwise. Projecting the latent back toward its level set during imagination reduces drift in *decoded physical energy* by 55–76% at long horizon on trajectories the method never saw, with 0 of 60 magnitude-matched random directions and 0 of 15 equal-norm tangent directions beating it. Setting the scalar to an independent trajectory's value moves the imagined world's true energy toward that trajectory's true energy (rank correlation 0.80–0.91, 0 of 75 controls). The same interchange applied 50 steps into an autonomous rollout — on a state the model reached by itself — works nearly as well, which is what a dormant pathway would not do [Makelov et al., 2024].

And it has boundaries worth stating. Outside the training energy band a freely fitted probe still recovers energy at 0.999 while the transition conserves it 55–$267\times$ worse: the representation generalises, the conservation does not. On a deterministic conv-GRU world model, identification transfers on 2 of 3 seeds and conservation on none, and training its transition seven times longer does not close the gap. The supportable scope is RSSM-like models, in distribution.

Contributions.

1. **A demonstration that probing over-reports, with a decisive control.** A supervised probe of the true physical quantity, at $|\rho| = 0.9999$, identifies a direction the dynamics do not use and that harms rollouts when enforced. The dissociation appears on five further independent axes and is directional in all of them.

2. **An operator statistic that discriminates where probes cannot**, together with the matched controls — magnitude-matched rather than norm-matched — that make its intervention interpretable. We show the norm-matched null used in prior work on this system controls nothing, because the correction is scale-invariant in the constraint.

3. **Causal evidence that the selected direction is deployed**, including an interchange 50 steps into autonomous imagination, and **a boundary condition**: the phenomenon is bounded to the training distribution and does not transfer to a deterministic recurrent architecture.

2 Recovering a latent invariant

Model and data. We analyze the published DreamerV3 architecture [Hafner et al., 2023]: 13.5M parameters, categorical 32×32 stochastic latents, a convolutional encoder and decoder, Kullback–Leibler balancing, and unimix. We train only the world model, with no actor or critic.

The data come from Gymnasium's `Pendulum-v1` [Towers et al., 2024]. Each simulator step renders a 500×500 frame, which we crop to 448×448 and block-average to 64×64. We set the simulator state directly, sampling $\theta \sim U[-1.8, 1.8]$ and $\dot{\theta} \sim U[-2.2, 2.2]$, and reject any trajectory that reaches the simulator's $|\dot{\theta}| = 8$ speed clip, since clipping would break conservation. Actions are always zero. A single frame does not reveal velocity, so the model must infer $\dot{\theta}$ from the sequence.

We train on 204 trajectories of 120 frames using Adam at 10^{-4}, batch size 16, and sequence length 64, capped at 30 minutes of wall-clock time, which gives about 6,500 gradient steps. The remaining 52 trajectories are reserved as a post-training *analysis set*. Three independently trained seeds pass checks fixed before analysis: KL divergence above 1 nat, one-step decoding at least 4× better than predicting the dataset mean, and finite rollouts. Appendix A gives the remaining training details. Our claims concern a DreamerV3 world model trained on pendulum video, not DreamerV3 as a full reinforcement-learning agent.

Latent coordinates. After training we freeze the model. Let h denote the deterministic recurrent state on the analysis set and $\bar{h}$ its mean. We take the top 12 principal directions of h, remove any direction with no support in the data, and let P denote the resulting orthonormal map. All extraction, projection, and perturbation operate on

$$z = P^\top (h - \bar{h}). \tag{1}$$

Let $F(z)$ be the projection through P of the model's one-step displacement $T(h) - h$. Using the model's own displacement avoids fitting a separate vector field and then analyzing that surrogate.

The candidate family. We look for the invariant among degree-4 polynomials in z. Writing $\varphi(z)$ for the vector of monomials up to degree 4 in the 12 latent coordinates, a candidate is $C(z) = a^\top \varphi(z)$, and the search is over the coefficient vector a. There are 1819 such monomials, from z_1 through $z_1 z_2 z_3 z_4$, excluding the constant term, which is conserved trivially. The pendulum's energy is quadratic in $\dot{\theta}$ and needs polynomial terms to approximate $\cos\theta$, and the latent is an unknown nonlinear encoding of $(\theta, \dot{\theta})$ rather than those coordinates themselves, so the family must be rich enough to express energy through that distortion while staying small enough to fit.

The invariance criterion. An invariant is constant along any one trajectory and differs between trajectories. We score a candidate by

$$\text{ratio}(C) \;=\; \frac{\text{mean within-trajectory variance of } C}{\text{total variance of } C}, \tag{2}$$

which is 0 for a perfectly conserved scalar and near 1 for one that wanders as much within a trajectory as across the dataset. The denominator keeps the criterion non-trivial: without it, $C = 0$ would score perfectly while distinguishing nothing.

Both variances are quadratic forms in a. With W the mean within-trajectory covariance of the features and T their total covariance, Equation 2 is $a^\top W a / a^\top T a$, a Rayleigh quotient. Its minimiser is the eigenvector of the generalized problem $Wa = \lambda T a$ with the smallest eigenvalue, and λ is the invariance ratio itself. One eigendecomposition returns the whole family, ranked from most to least conserved.

Selecting among conserved candidates. Conservation alone does not pick out a unique scalar. If C is conserved then so is any function of it, so E, E^2 and mixtures of E with other conserved quantities all sit near the top of the ranking, and the leading eigenvector is generally some combination of them. We therefore use flow alignment as a secondary criterion among the leading candidates, fitting C jointly with an antisymmetric operator B such that

$$F \approx B \nabla C, \qquad B^\top = -B. \tag{3}$$

This mirrors the Hamiltonian relation between a conserved quantity and the flow it generates: an antisymmetric operator maps the gradient of C to a direction tangent to its level set [Arnold, 1989]. Because $\nabla C^\top B \nabla C = 0$ for any antisymmetric B, a scalar satisfying Equation 3 is constant along the flow by construction. The criterion also separates candidates the invariance ratio cannot: E^2 is exactly as conserved as E but does not pair with the same B.

The fit is bilinear, so fixing either a or B makes the other a least-squares problem and we alternate. It searches within the top eight eigenvectors from Equation 2, so the recovered C has eight effective degrees of freedom rather than 1819. That matters for a search run on 52 trajectories, where a free fit over the full basis could drive the in-sample ratio to zero by overfitting. Appendix A isolates the contribution of the flow criterion: conservation drives most of the recovery and intervention effect, while flow alignment mainly reduces variation across seeds.

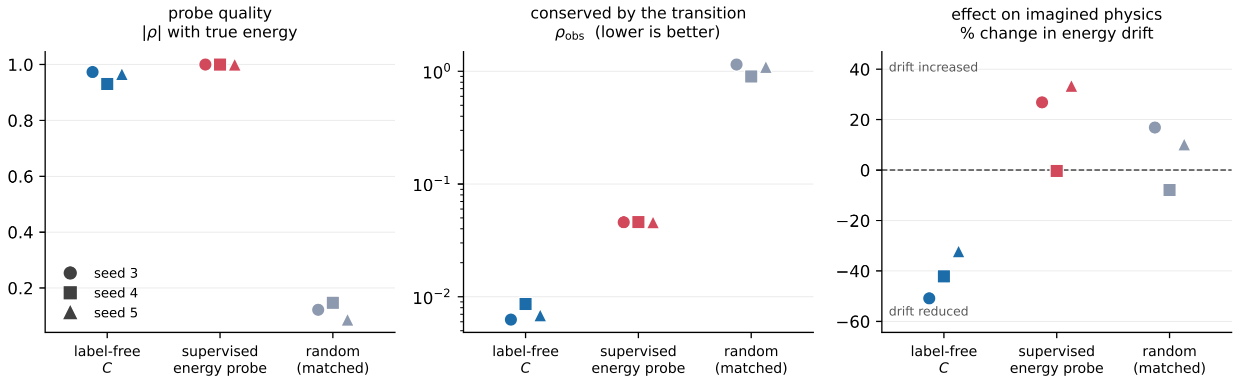

Figure 1: A perfect probe is not a conserved quantity. Colour is the constraint; marker is the model seed (all three shown, never averaged). **Left:** the supervised probe reads true energy almost perfectly. **Centre:** the model's own transition preserves it an order of magnitude worse than the label-free scalar. **Right:** acting on it fails to reduce rollout energy drift, while the label-free scalar cuts drift on every seed. Probe quality and dynamical relevance rank the three constraints differently.

Reference energy. We compare C with the textbook pendulum energy $E = \frac{1}{6}\dot{\theta}^2 + 5\cos\theta$. Gymnasium uses a semi-implicit integrator, so it conserves a nearby shadow Hamiltonian $\tilde{H} = H + O(\Delta t)$ rather than E itself [Hairer et al., 2006a]. In our trajectories this produces about 12% relative oscillation in E without secular drift, which we treat as a noise floor.

The search sees only latent states and the model's own one-step transition. It never sees θ, $\dot{\theta}$ or E, and nothing in Equations 2 and 3 refers to them. Ground truth enters afterward, when we score the recovered C against E. We selected the extraction dimension on three development models and fixed it before evaluating the three models reported here.

3 Decodability is not dynamical structure

A probe establishes that a quantity can be read out of a representation. It says nothing about whether the model's own transition uses or preserves that quantity. We measure the second property directly, with a statistic on the transition operator: apply one autonomous step at a real encoded state and ask how much a candidate scalar changes,

$$\rho_{\rm obs}(C) \;=\; \frac{\mathrm{median}_t \big| C(T(z_t)) - C(z_t) \big|}{\mathrm{std}_{\rm traj}\, C}, \tag{4}$$

normalised so it is invariant to the scale of C and comparable across candidates. A value near zero means the transition preserves C; a value near one means a single step moves C by its entire across-trajectory spread.

A perfect probe identifies a direction the dynamics do not use. The sharpest test replaces the label-free search with the answer it is trying to find. We fit a degree-4 polynomial readout to *true* pendulum energy by ridge regression on the same latent, then use it to drive the identical correction. It reaches $|\rho|_E = 0.9999$: as a probe it is essentially perfect.

Table 1 and Figure 1 show the outcome. The supervised probe tracks energy better than the label-free scalar on every seed, and is 6.7× less preserved by the transition (5.3–7.3× per seed); both hold on 3 of 3 models. It also never repairs the rollout: its effect on energy drift is +26.8% and +33.4% on two models and −0.3% on the third, against a 32–51% reduction from the label-free scalar on all three. The direction that best predicts the physical variable and the direction the model's dynamics actually respect are not the same direction.

This also answers the practical objection to a label-free method. The search is not a workaround for missing labels: it finds something a supervised fit on perfect labels does not, because it optimises conservation

Table 1: The correct physical quantity, fitted optimally, is worse conserved and does not repair rollouts. Median across three independently trained models, with the per-seed range in brackets; 20 magnitude-matched random constraints per model (60 total, pooled).

constraint	$\lvert\rho\rvert_E$	ρ_{obs}	change in energy drift
label-free C	0.966 [0.930, 0.973]	$\mathbf{6.9}\,[\mathbf{6.3,\ 8.7}]\times10^{-3}$	**−42.2% [−50.9, −32.2]**
supervised energy probe	**0.9999**	$4.58\,[4.57,\ 4.59]\times10^{-2}$	+26.8% [−0.3, +33.4]
random, magnitude-matched	0.126	1.07	+4.0% [−20.7, +42.6]

Table 2: Decodability is retained while dynamical structure degrades, across five independent axes. ρ_{obs} for the trained conservative model in-distribution is 6.3–8.7×10^{-3}.

axis	contrast	decodability	ρ_{obs} degradation
training	untrained network	$\lvert\rho\rvert_E$ up to 0.908	74×
state space	outside training energies	free probe 0.999	55–267×
architecture	deterministic conv-GRU	$\lvert\rho\rvert_E$ 0.91 (2 of 3 seeds)	767×
objective	transition never trained	$\lvert\rho\rvert_E$ 0.97	∼10 000×
actuation	torque applied	$\lvert\rho\rvert_E$ 0.72–0.88	balance law absent

by the transition rather than agreement with the variable. Section 4 identifies what it finds, and why fitting true energy cannot find it.

The dissociation is directional, and appears on five further axes. In every case we can construct, decodability survives where conservation fails; we have not found the converse. Table 2 collects them. On each row a probe would report success.

The out-of-distribution row is the cleanest, because it holds the model fixed and varies only the region of state space. Outside the training energy band a freely fitted probe recovers energy at 0.999 — the encoder generalises — while the transition conserves it 55–267× worse, reaching values statistically indistinguishable from a random constraint at the earliest checkpoint. Decodability and conservation come apart inside a single trained model, split by where in state space it is asked.

Under actuation the same split appears against a different law. The four axes above all vary how well a *conserved* quantity survives. A fifth varies the law itself. We train three further world models on the same pendulum under piecewise-constant random torque, where energy is no longer conserved but obeys a balance relation, $\mathrm{d}E/\mathrm{d}t = \tau\dot{\theta}$. The models use their actions: a 20-step open-loop rollout driven by the true action sequence beats one driven by the same actions shuffled across trajectories, at a ratio of 0.740–0.761 on 3 of 3 seeds.

Running the *unchanged* extraction on these models recovers an energy-like scalar on every one, $\lvert\rho\rvert_E = 0.72$, 0.88, 0.86. Its motion under the model's own transition is then compared against the balance relation the physics demands, with the torque known and $\dot{\theta}$ taken from the same decoded readout used throughout. The rank correlation between the predicted source term and the observed change in C is 0.041, 0.076 and 0.065; the power term accounts for **0.31%** of the variance in ΔC, and the observed change is 4.5–5.4× larger than the relation predicts. A control that shuffles the power term across transitions is beaten on 3 of 3 seeds, so the pairing carries something — but almost nothing.

The model represents the quantity and does not implement the relation that quantity obeys. This is the same dissociation as above, stated against a balance law rather than a conservation law, and it is the reason a correction built on the recovered scalar has no control benefit (Section 6): there is no balance structure in the latent for it to enforce.

Whether conservation *alone* predicts the intervention is not established. The control that would show this directly asks whether, among candidates matched in energy correlation, better conservation alone produces a larger repair. The 2-DoF system admits such a band: on two of three seeds, 397 and 398 of 400 candidates lie within 0.10 of the recovered scalar's $\lvert\rho\rvert_E$, spanning five orders of magnitude of conservation quality at $|\rho|_E \leq 0.11$. Across 20 candidates stratified by conservation, the rank correlation with repair is $+0.19$, $+0.57$ and $+0.19$ on the three seeds, with the 95% confidence interval excluding zero on **one of three**. We report this as a negative result: the relationship is visible but not established at $n = 3$.

Two further facts belong with it. The point estimate moves with the size of the candidate pool ($+0.29 \to +0.19$ on one seed between pools of 150 and 400), so the statistic is not stable to a design choice that should be immaterial — though the one-of-three verdict is identical under both pools. And on the third seed the band cannot be built at all: its reference $|\rho|_E$ is 0.111 and only 19 of 400 candidates fall within 0.10 of it, with an increase from 150 to 400 candidates adding none.

That last failure is the informative part. On a system whose transition conserves energy, the well-conserved latent directions *are* the energy-like ones. This is why E10's original high-decodability form could not be constructed on the pendulum at any pool size, why the matched band here exists only at low decodability, and why on one seed it does not exist at all. The two properties may not be separable by this design on any system where the model has learned the physics. That is a statement about the difficulty of *this control*, not about the dissociation itself, for which the supervised probe above and Section 4 give direct evidence. What does hold across all three seeds is that the recovered scalar repairs: -16.7%, -7.3% and -6.8%.

Where the phenomenon stops. Both boundaries above are the same statement seen twice. The invariant is bounded to the *training distribution*: outside the training energy band a freely fitted probe still recovers energy at 0.999 while the transition conserves it 55–267× worse, reaching values indistinguishable from a random constraint. And it is bounded to the *architecture*: on a deterministic convolutional GRU with no stochastic latent, no KL term and no unimix, identification transfers on 2 of 3 seeds under both training budgets while conservation transfers on none (ρ_{obs} 4.83–5.55 against the RSSM's $\sim 7\times 10^{-3}$, a 767× gap; seven times more open-loop training moved the median from 5.33 to 5.26). We are explicit that this comparison does not isolate architecture — an open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths — so the supportable scope of every claim here is RSSM-like models, in distribution.

4 Why the perfect probe fails

Section 3 shows that the optimal readout of true energy is a poor invariant of the model's transition. It does not say why. This section gives the reason, and the reason turns out to be that the probe is fitted to the wrong quantity.

The data does not conserve textbook energy. The trajectories are generated by semi-implicit (symplectic) Euler,

$$\dot\theta_{n+1} = \dot\theta_n + \tfrac{3g}{2l}\sin(\theta_n)\,\Delta t, \qquad \theta_{n+1} = \theta_n + \dot\theta_{n+1}\,\Delta t. \tag{5}$$

A symplectic integrator does not conserve the Hamiltonian H of the continuous system. It conserves a nearby *shadow* Hamiltonian $\tilde H = H + \mathcal{O}(\Delta t)$ [Hairer et al., 2006a]. For this system the first-order correction is $\frac{\Delta t}{2}\{T, V\} \propto \dot\theta \sin\theta$, giving

$$\tilde H \;=\; E \;+\; c^\star\,\dot\theta\sin\theta, \qquad c^\star \;=\; \tfrac{\Delta t}{2}\,mg\tfrac{l}{2} \;=\; 0.125, \tag{6}$$

with no free parameters. If the model has learned the integrator's map rather than the textbook physics, then $\tilde H$, not E, is what its transition should preserve — and a probe fitted to E is fitting the wrong target however well it fits.

A sweep, not a two-point test. Comparing a probe fitted to E against one fitted to $\tilde H$ would not settle this: $\tilde H$ is a different function, and might be better conserved merely because it is easier to represent in the degree-4 basis. We therefore sweep the whole family

$$T_c(\theta,\dot\theta) \;=\; E(\theta,\dot\theta) \;+\; c\,\dot\theta\sin\theta, \tag{7}$$

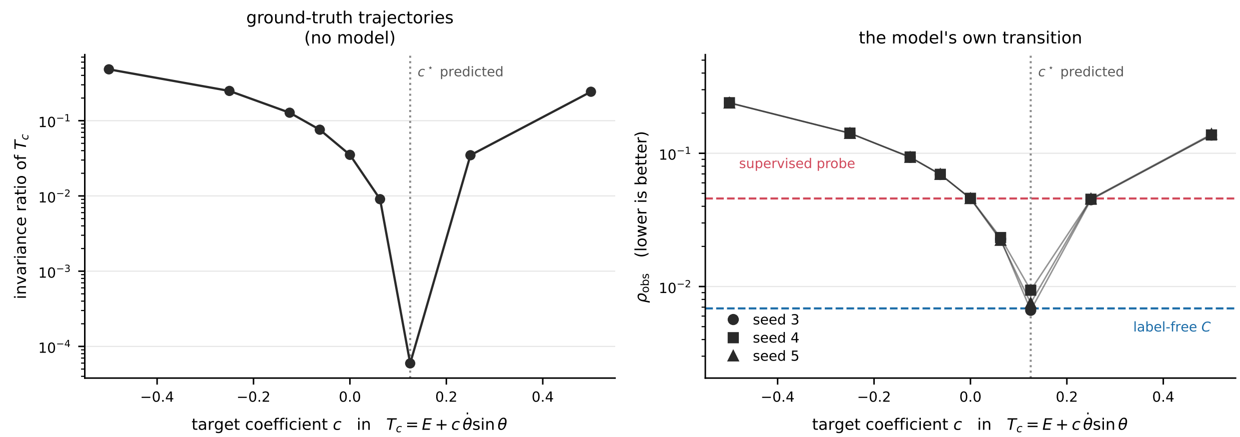

Figure 2: The model's transition conserves the integrator's shadow Hamiltonian, not textbook energy. Both panels sweep the target $T_c = E + c\,\dot\theta\sin\theta$; the dotted line is $c^\star = 0.125$ predicted from the integrator with no fitting. **Left:** on ground-truth trajectories, with no model involved, $T_{c^\star}$ is 591× better conserved than textbook energy. **Right:** the learned transition agrees, on all three seeds. The sweep passes from the supervised probe's level (red) down to the label-free scalar's (blue) exactly at $c^\star$. The wrong-sign control at $c = -c^\star$ is 12× worse (median over seeds), so this is not monotone in $|c|$.

Table 3: The recovered correction tracks the simulator's timestep across a 4× range. Three models per timestep; ρ_{obs} is the median across seeds.

Δt	predicted $c^\star$	$\arg\min_r$ per seed	ρ_{obs} at r=0	at r=1	separation
0.02	0.050	+0.75, +0.75, +1.00	0.0171	0.0166	1.03×
0.035	0.088	+1.00 × 3	0.0231	0.0103	2.24×
0.05	0.125	+1.00 × 3	0.0449	0.0079	5.72×
0.08	0.200	+1.00 × 3	0.1188	0.0088	**13.5**×

over a grid containing $c^\star$, *both signs*, and wrong magnitudes at 2× and 4× either way, refitting the probe at each c and holding everything else fixed. At $c = 0$ the procedure is exactly the supervised probe of Section 3, which reproduces its numbers to five decimal places. The prediction is then the *location of a minimum*, which no representational-ease artifact produces. It was registered before the sweep was run.

The minimum is where the integrator says it is. Figure 2 shows both halves. On ground-truth trajectories, with no model involved, the invariance ratio of T_c falls by 591× between $c = 0$ and $c = c^\star$, confirming the derivation and fixing the sign. On the learned transition, ρ_{obs} is minimised at $c^\star$ on 3 of 3 models, falling 4.9–6.9× below its value at $c = 0$. The curve is not monotone in $|c|$: the wrong-sign, right-magnitude control at $c = -c^\star$ is 12× worse than at $c = +c^\star$ (range 10–14× over the three models). A post-hoc refinement over $c \in [0.090, 0.170]$ locates the minimum at 0.125 ± 0.005 on every seed.

The repair follows the same curve. Enforcing $T_{c^\star}$ during imagination changes energy drift by −49.5%, −31.5% and −29.3% on the three models — a repair on all three — where enforcing T_0, the perfect probe, gives +26.8%, −0.3% and +33.4%. Correcting an $\mathcal{O}(\Delta t)$ term in the target flips the intervention from harmful to helpful.

The model tracks the simulator's timestep. If the model has learned the integrator rather than the physics, then $c^\star$ is not a constant of the pendulum but a property of the discretisation, and changing the simulator's timestep should move it: $c^\star(\Delta t) = \frac{\Delta t}{2} mg \frac{l}{2} = 2.5\,\Delta t$. We test this directly. Four datasets are generated at $\Delta t \in \{0.02, 0.035, 0.05, 0.08\}$, identical in every other respect, and three models are trained on each. The sweep is expressed relatively, $r = c/2.5\Delta t$, so the prediction is a single value of r at every timestep.

Table 3 shows the outcome. The minimum sits exactly at the predicted $r = 1$ on 10 of 12 models, and the wrong-sign control is beaten on 12 of 12. Regressing the recovered coefficient on Δt through the origin, as the theory requires, gives a slope of 2.484 ± 0.058 against a parameter-free prediction of 2.500 — agreement to 0.6% with nothing fitted.

Two scalings make the mechanism visible. As the timestep coarsens, ρ_{obs} for textbook energy grows as $\Delta t^{1.41}$, while ρ_{obs} for the shadow quantity stays at the model's own floor, 0.008–0.017 across the whole range. The separation between them therefore grows from 1.03× to 13.5× purely because textbook energy becomes worse conserved while the shadow quantity does not. **At every timestep, the model conserves the integrator's quantity rather than the physicist's.**

We report one registered failure. Our preregistration required both a slope within 20% and an intercept whose interval contains zero; the two-parameter fit gives a slope of 2.611 ± 0.110 but an intercept of -0.0072 ± 0.0057, which excludes zero. It is driven entirely by the smallest timestep, where a post-hoc finer grid puts the minimum at $r = 0.875$ on all three seeds — a real 12% low bias rather than grid coarseness, in the regime where the whole separation is only 1.03× and the residual curve is flat to within 2%.

What this does and does not settle. The residual matters and we report it: the median ρ_{obs} at $c^\star$ is 7.5×10^{-3} against the label-free scalar's 6.9×10^{-3}, a remaining gap of 1.10×. The shadow Hamiltonian therefore accounts for essentially all of the 6.7× gap in Section 3. It does *not* follow that the model conserves $\tilde{H}$ exactly: the model is a learned approximation to the integrator, so its own invariant is its own modified quantity, of which $\tilde{H}$ is a first-order proxy. That 1.10× is the measure of the difference.

Two consequences are worth stating plainly. First, a probe reaching $|\rho|_E = 0.9999$ can be fitted to a quantity *the generating process does not conserve*, and no amount of probe hygiene detects this, because nothing is wrong with the probe — the target is wrong. Second, the label-free method's advantage now has a name. It optimises conservation under the learned transition, so it is not restricted to targets a human knows how to write down; here it recovered an $\mathcal{O}(\Delta t)$ integrator correction that nobody supplied. The sharpest statement of this paper's thesis is internal to a single sweep: at $c^\star$ the probe correlates with textbook energy *worse* than at $c = 0$ ($|\rho|_E$ 0.977 against 0.9999), while being conserved six times better and repairing the rollout instead of harming it.

5 The privileged direction is causally deployed

If the recovered scalar is only a descriptive correlate, enforcing it has no reason to help and setting it has no reason to steer. Three interventions test the alternative, each with matched controls.

A magnitude-matched null, and why the obvious one fails. The natural correction projects the latent back toward the level set, $z \leftarrow z - \alpha(C(z) - C_0)\,\nabla C/\|\nabla C\|^2$. This is a Newton step, so its size grows with how badly the constraint is violated — and a random constraint is violated far more than a conserved one. Matching random draws in *coefficient* norm, as one might expect to control for scale, controls nothing: the step is invariant under $C \to \lambda C$. Measured, random draws take steps 29× larger than the recovered constraint, so the comparison conflates direction with magnitude.

We therefore fix the step size and vary only direction, $z \leftarrow z - \varepsilon\,\text{sign}(C - C_0)\,\nabla C/\|\nabla C\|$, over a frozen ε grid. Arms then differ in where they push, never in how hard.

Correcting the invariant repairs the imagined physics. We evaluate on decoded physical energy, not pixels: a non-learned geometric readout recovers θ from the rendered frame by image moments and $\dot{\theta}$ by backward differences, matching the simulator's semi-implicit convention, with a median decoded-energy error of 1.6% of the across-trajectory spread. The statistic is the secular drift of decoded energy over a rollout.

At $H = 100$ the recovered direction reduces drift by 32–51% across three models. On 512 trajectories from a disjoint generator seed, with C and the entire coordinate frame frozen, the effect does not shrink: −54.6%, −58.8%, −75.9% at $H = 190$. Across both horizons **0 of 60** magnitude-matched random directions and **0 of 15** equal-norm tangent directions beat the recovered one on any model.

Setting the invariant steers the imagined physics. Restoring C shows drift is costly; it does not show C is a control variable. We take a recipient state and an independent donor trajectory, move the recipient minimally along the local C-normal until $C(z) = C(z_{\text{donor}})$, and let the rollout run free. The intervention never sees true energy — the target is read from the model's own latent.

The imagined world's *true decoded* energy moves toward the donor trajectory's *true* energy, with rank correlation 0.802–0.914 across three models and **0 of 75** random and tangent controls beating it. A preregistered offset sweep is monotone in both directions on every model.

The subspace is used by the model's own dynamics, not only reachable from encoder states. A subspace edit can produce the expected output change through a pathway the model does not normally use [Makelov et al., 2024]. We therefore repeat the donor interchange *50 steps into an autonomous rollout*, on a state the model reached by itself rather than one the encoder produced.

The effect barely degrades: rank correlation 0.762–0.849 at depth 50 against 0.803–0.924 at depth 0, with 0 of 13 controls beating it at either depth on any model. Equal-norm tangent edits, which cannot change C to first order, sit near zero at both depths. A dormant pathway would lose its effect precisely where this test applies it.

The same pipeline refuses when there is nothing to find. On matched dissipative models, where no nontrivial first integral exists, the extraction returns a direction that is not distinguished from a random one: 24 of 60 random draws beat it, and it harms rollouts on 2 of 3 models. Its operator statistic sits at $3.4\text{–}3.7 \times 10^{-1}$, in the same range as an untrained network and 39× worse than any conservative model, with no overlap between the two groups.

6 Limitations

Two physical systems, both smooth and Hamiltonian: a pendulum and a two-degree-of-freedom anharmonic oscillator. No contacts, no multiple interacting objects, no actions. Whether any of this survives discontinuous dynamics is untested.

Three trained models per arm. We report ranges rather than confidence intervals for model-level statistics throughout, because three is too few for a precise effect size, and we take the model seed rather than the trajectory as the independent unit for every model-level claim.

The extraction searches a degree-4 polynomial family in a 12-dimensional principal subspace. The invariance ratio it minimises *is* the normal component of one-step error, so the recovered scalar having low normal error is the objective being met rather than a finding; the empirical content is that the resulting direction coincides with physical energy, which Section 3 shows supervision does not reproduce.

The architecture comparison does not isolate architecture. An open-loop reconstruction term and a prior–posterior KL are different losses, not the same loss at different strengths, so the conv-GRU result narrows the explanation toward the latent design without establishing it.

We show no downstream consequence, and both uses we tested fail — for the same measurable reason.

Whether the correction helps a planner was tested directly. We plan with CEM over the learned world model on an energy-targeting task inside the training distribution, scoring return from the simulator's true state. The model plans well: it beats a random-action policy by +1.65 in mean return, 95% CI [+0.50, +2.79]. But no correction arm — enforcing the recovered scalar, enforcing it with the action's power supplied as a source term, enforcing the supervised probe, or a magnitude-matched random direction — differs from no correction at all, over three models and 20 episodes each. The largest arm effect is 0.0014 against an across-episode return standard deviation of 0.575: **0.24% of one episode's spread**. All four registered predictions failed, including the prediction that the label-free scalar would beat the supervised probe.

The reason is measured rather than inferred, and it is structural. The correction acts on *accumulated* drift: at the planner's horizon it shifts imagined energy by 0.3% of the spread across candidate action sequences, which is the quantity the planner ranks plans by, so it cannot change which plan is chosen. Our repair result is measured over 100 autonomous steps; a controller that re-plans every step never imagines long enough to accumulate the drift the method removes. This bounds where the method can matter — long open-loop imagination, not closed-loop control — and it is the same effect-size limit that defeats the trust signal below.

Planning further ahead does not recover the effect, and the reason is that the two requirements are in opposition. Tripling the planning horizon leaves every arm still indistinguishable from no correction (largest effect 0.007 against an episode return standard deviation of 0.862), while planning quality itself *degrades*, from -0.34 to -0.56 in mean return: the model's rollout error grows with horizon faster than the value of looking further ahead. The regime in which the correction would have something to correct is the regime in which the planner is already worse off.

Whether invariant drift predicts rollout failure was tested directly, and it does not: accumulated drift at rollout step 25 correlates with decoded energy error at step 100 at Spearman -0.02, $+0.09$ and -0.24 across three models, against $+0.27$, $+0.29$ and $+0.31$ for plain latent displacement $\|z_k - z_0\|$, with a random-constraint control beating it on two of three. The registered prediction was that drift would beat displacement; the result went the other way on every model. A conserved quantity being causally deployed in imagination therefore does not imply that its drift is a useful online signal, and we make no such claim. The effect on pixel error alone at short horizon is also small (1.7% at $H = 50$); the large effects reported here are on decoded physical energy, a metric we introduce because it measures the property in question.

The method also finds only what the data gave the model a reason to represent. We tested this on a constraint the pendulum satisfies exactly — its rod length is fixed, so the rendered bob's distance from the pivot is constant to 0.67% — by searching for a residual $G(z) = 0$ rather than a conserved scalar. The search fails: the recovered direction correlates with the true residual at 0.004–0.032 across three models. Two causes, both measured. Minimising $\mathbb{E}[G^2]$ is the wrong objective, since 1525 of 1819 candidate directions have a smaller second moment than the true constraint. And the constraint is barely encoded at all: predicting the rod length from the latent gives held-out $|\rho| = 0.19$, with a residual four times larger than the quantity's own variation. A quantity that never varies carries no information, so nothing in a reconstruction objective pushes it into the latent — it can live in the decoder's weights. This bounds the approach to quantities that vary, and it is why we do not report a released-checkpoint experiment: limb lengths do not vary either.

Finally, the two-invariant case exposes a limitation of the extraction rather than of the model. On a system conserving both energy and angular momentum, the conserved subspace reliably contains both (angular-momentum correlation 0.94–0.98 in every measurement) while the joint fit isolates either in only 1 of 7 measurements across three models. The flow-alignment criterion, which selects a single direction inside the conserved subspace, has no unique answer when two independent invariants exist.

7 Related work

What probes establish. Probing shows a variable is recoverable from a representation, not that the model's computation uses it [Alain and Bengio, 2017]. Control tasks [Hewitt and Liang, 2019] and later critiques [Belinkov, 2022] address the confound between what the representation stores and what the probe can learn. Our result is a different failure: the probe in Section 3 is fitted to the true physical quantity and reaches 0.9999, so no control-task correction applies to it, and it still identifies a direction the dynamics do not use. Work on emergent world representations goes beyond probing by intervening [Li et al., 2023, Nanda et al., 2023], which is the logic our interventions follow; Makelov et al. [2024] show such interventions can succeed through pathways the model does not normally use, which is the objection our depth-50 interchange is designed to answer.

Discovering conserved quantities. A substantial literature finds conservation laws from data. ConservNet minimises a noise-regularised within-group variance [Ha and Jeong, 2021], of which our invariance ratio is the closed-form Rayleigh-quotient analogue; AI Poincaré counts conserved quantities from trajectory manifolds [Liu and Tegmark, 2021]. ConCerNet learns a conservation law and then projects the learned dynamics onto it [Zhang et al., 2023], and FINDE finds first integrals and preserves them by projection [Matsubara and Yaguchi, 2023]; Noether Networks meta-learn a conserved quantity and apply it at test time, including on pixel pendulums [Alet et al., 2021]. All of these *impose or co-train* conservation. We ask a different question: whether a model trained with no such term has a scalar its transition already preserves, and we measure that on the frozen transition rather than building it in.

Structure by design, and whether it is what helps. Hamiltonian and Lagrangian networks build conservation into an architecture [Greydanus et al., 2019, Cranmer et al., 2020]. Gruver et al. [2022] find that HNN generalisation comes from modelling acceleration and coordinate choice *rather than* from symplectic structure or energy conservation — which licenses the objection that our projection is a generic regulariser. Our magnitude-matched null answers it directly: with step size fixed, 0 of 60 random directions and 0 of 15 equal-norm tangent directions match the recovered one, and on dissipative models the same pipeline returns a direction indistinguishable from random.

Drift in learned dynamics. That accurate one-step models drift over long rollouts is documented across fields. In machine-learned force fields, force accuracy and simulation stability decouple sharply [Fu et al., 2023]. In numerical analysis, projection onto a level set is a standard structure-preserving device [Hairer et al., 2006b], and our correction is that device applied to a *discovered* rather than a given invariant, in a learned latent rather than the physical state.

Evaluating physics in world models. World models are normally evaluated by prediction quality or control performance [Hafner et al., 2023, Ha and Schmidhuber, 2018, Schrittwieser et al., 2020], and physics benchmarks evaluate generated output. Neither reads the model's internal state. The disentanglement literature [Locatello et al., 2019] documents how readily an appealing structural claim about representations survives without the control that would falsify it.

8 Conclusion

A probe fitted to the correct physical quantity, at a correlation of 0.9999, identifies a direction a world model's dynamics do not use — and enforcing it degrades the model's imagined physics, while a label-free scalar that is better conserved and slightly less accurate improves it. Decodability and dynamical structure are different properties of a representation, they come apart in every setting we could construct, and only a measurement on the transition operator tells them apart.

The gap has a cause, and it is not a defect of the probe. The training data is produced by a symplectic integrator, which conserves a shadow Hamiltonian $\tilde{H} = H + \mathcal{O}(\Delta t)$ rather than H itself. Sweeping the regression target through a family containing both signs and wrong magnitudes, the direction the learned transition conserves best sits at exactly the coefficient the integrator predicts — no free parameters, 3 of 3 models — and correcting that $\mathcal{O}(\Delta t)$ term turns a harmful intervention into a helpful one on every model. The perfect probe was fitted to a quantity the generating process does not conserve. No amount of probe hygiene detects that, because nothing is wrong with the probe. This also names what the label-free search buys: it optimises conservation under the learned transition, so it is not confined to targets a human can write down, and here it recovered an integrator correction nobody supplied.

The practical consequence is narrow and concrete. Finding that a physical variable is decodable from a world model's latent says little about whether the model's rollouts respect it, and it says least in the situations that matter most: outside the training distribution, in architectures that predict well without conserving, and when the variable is supplied by supervision rather than found in the dynamics. Reporting a probe result in those settings will overstate what the model has learned. The operator statistic costs one autonomous step per state to compute.

A Experimental details and extraction ablations

Three kinds of latent trajectories. We use three latent trajectories and keep them separate throughout.

An *observation-conditioned* trajectory is the sequence of deterministic recurrent states produced when the model receives the next real frame at every step. This is what the encoder returns on the analysis set.

The *one-step transition* T is the model's autonomous recurrent update with zero action and no new observation.

An *imagination rollout* starts from one encoded state and repeatedly applies T without supplying any later frames.

We estimate the latent flow by applying T once at each state of an observation-conditioned trajectory, so the transition is autonomous even though the states come from real video. Section 3 measures conservation along observation-conditioned trajectories. Section 5 tests what happens to the same scalar under autonomous imagination.

Code and data. The extraction code, the run logs behind every figure and table, and the source of this paper are at `https://github.com/Zarand3r/world-model-invariants`. The figures regenerate from the committed logs without a GPU.

Preliminary experiments. We developed the extraction procedure in preliminary experiments on smaller recurrent models. Those experiments motivated the untrained, dissipative, and intervention controls used here, and we do not use them as evidence for the DreamerV3 results.

Training. We set free bits to 0 rather than DreamerV3's reference default of 1.0. The measured KL divergence settles near 1.8 nats, well above collapse, so the floor is inactive here. All three trained models pass the acceptance checks; none were discarded.

Data split. Training uses trajectories 0–203 of the 256-trajectory dataset. The remaining 52 form the analysis set. We fit C on those 52 trajectories and evaluate the Section 5 correction on the same 52, so the absolute effect is in-sample with respect to invariant fitting. The recovered and random-constraint arms use the same trajectories.

Scoring the correction over a grid. We score the slope of rollout error over the fixed α grid rather than the minimum on that grid. Taking the minimum selects the most favourable of five points after seeing the curve. In one diagnostic run that rule would have reported a 5.9% improvement for an arm whose error rose monotonically to twice baseline.

Flow alignment contributes reproducibility. Holding the candidate family, data, dimension, degree, and α grid fixed, we select C either by the invariance criterion alone or by the joint criterion. Conservation alone recovers energy at 0.950 against 0.973 for the joint rule, and gives a slightly better mean intervention slope (-0.087 against -0.077, winning on two of three models). The difference is in spread: recovery varies by 0.035 across seeds under conservation alone and by 0.008 under the joint rule.

The pairing residual does not track recovery. As the extraction dimension rises from 6 to 12, energy recovery improves from 0.721 to 0.967 while the pairing residual worsens from 0.738 to 0.876. Recovery already reaches 0.967 at dimension 8, where the residual is lower at 0.813. The added dimensions do not carry energy while resisting the Hamiltonian fit: directions 7–12 account for 44.9% of the residual and 43.5% of the flow. Figure 4 gives the sweep.

B Latent sensitivity analyses

For each direction u of the extracted subspace we measure its variance and the damage a displacement along it does to an imagination rollout,

$$V(u) = \mathrm{Var}(u^\top z), \qquad D_H(u) = \mathbb{E}\big[L_H(z + \epsilon u) - L_H(z)\big], \tag{8}$$

with L_H the H-step imagination error against the analysis set.

Calibrating the displacement. At $\epsilon = 0.05|z|$ the damage is about -0.5% of baseline with the sign varying across directions. Imagination is deterministic, so repeated unperturbed rollouts are identical and these values are not sampling noise. Damage for the leading direction runs -0.5%, $+3.1\%$, $+18.1\%$, $+46.4\%$, and $+128.8\%$ at $\epsilon/|z|$ of 0.05, 0.10, 0.25, 0.50, and 1.00. We use $\epsilon = 0.25|z|$, the smallest displacement whose damage exceeds 10% of baseline while the response still grows smoothly.

A displacement does not decay under imagination. After 40 steps it retains 6.3× its initial size at $\epsilon = 0.05|z|$, 2.8× at $0.25|z|$, and 1.9× at $0.5|z|$, as medians over the three models.

Variance and sensitivity at different horizons. At a 40-step horizon the two rankings anticorrelate on all three models (−0.364, −0.259, −0.406), with damage spanning 13.5× to 118.8× between the most and least consequential direction (Figure 3a). The ranking replicates: split-half rank correlation of D_H has median 0.95 across the settings we tried, with a minimum of 0.52 at the smallest displacement and longest horizon. The disagreement with variance does not extend to long horizons. Pooling three models and three displacements, median $\rho(V, D_H)$ is −0.252 at horizon 20, negative in 9 of 9 settings; −0.196 at horizon 50, negative in 5 of 9; and −0.028 at horizon 100, negative in 5 of 9, with one setting reaching +0.727. Once a rollout has diverged, high-amplitude directions dominate the error and the rankings realign.

The correction pushes along higher-damage directions at +0.385, +0.692, and +0.287 (Figure 3b), without having been given that ranking. The correlations stay well below 1, so the correction is only partly concentrated there. All of this is measured on the conservative models.

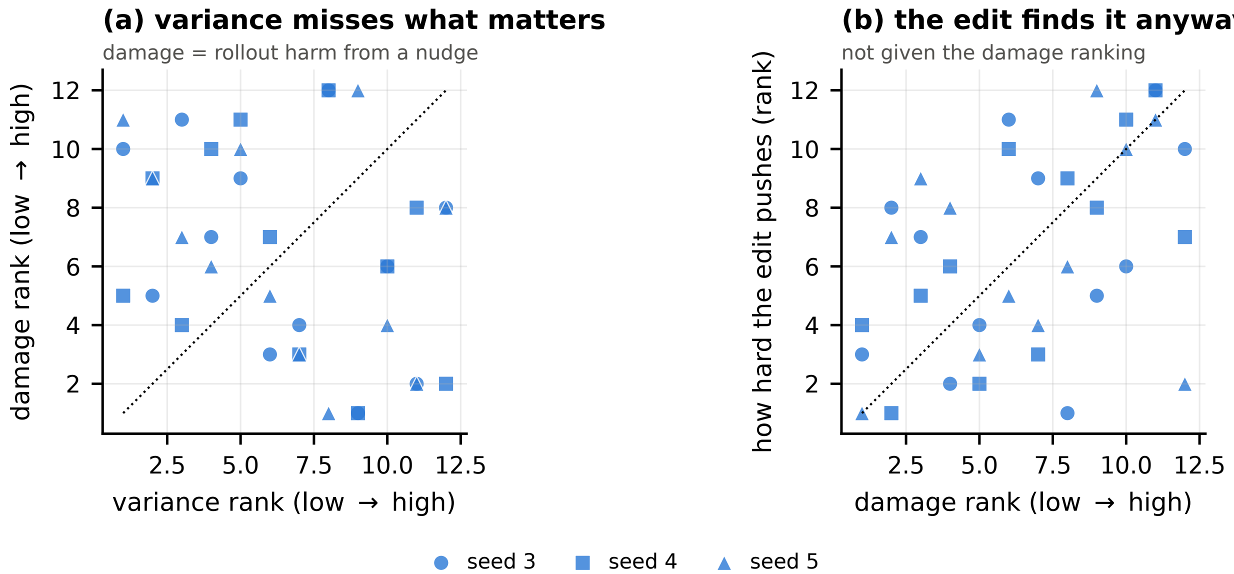

Figure 3: Each point is one of the 12 extracted directions, ranked within its own model; the dotted line marks equal ranks. **(a)** Variance rank against damage rank at a 40-step horizon. **(b)** Damage rank against the magnitude of the invariant correction along that direction.

Basis-independent properties reproduce better than coordinate selectors. The three models agree closely on quantities that need no choice of basis: energy recovery varies by 0.008, intervention slopes by 0.016, and the top-3 eigenvalue mass of $M_C = \mathbb{E}[gg^\top/\|g\|^2]$ with $g = \nabla C$ by 0.046 (0.448, 0.441, 0.487). Selectors on individual coordinates are less stable: ranking coordinates by ∇C and keeping the top three gives intervention slopes of −0.075, +0.016, and −0.089, beating a variance ranking on two models and changing sign across seeds. An orthogonal projector onto the top eigenspace of M_C beats the variance selector on all three models, though on one it sits at the 41st percentile of 100 random rank-3 subspaces. Since the correction is rank one at each state, the spectrum of M_C measures how much ∇C rotates as the state changes, and the amount is similar across models. This is consistent with the models encoding the same scalar in different latent coordinate systems, although we do not directly align their representations.

C Additional sweeps

Energy outside the highest-variance directions. On two of three models, principal directions 7–12 carry about twice the energy information of the top six (2.23× and 2.13×); on the third the ratio reverses to 0.91. We report the models separately because a median would hide the sign change. Figure 5 shows the decomposition.

Accumulated invariant error. Accumulated invariant error predicts which rollouts degrade most: an integral predictor reaches $R^2 = 0.187$ against 0.021 for a fitted power law in time at the horizon where both perform best, and shuffling destroys the relation. The current experiments do not distinguish weighted from unweighted accumulation. In a nested comparison with 400 bootstrap resamples, the weighted term adds explanatory power beyond the unweighted term in 1 of 9 seed–horizon combinations, and the unweighted beyond the weighted in 0 of 9, at $n = 512$.

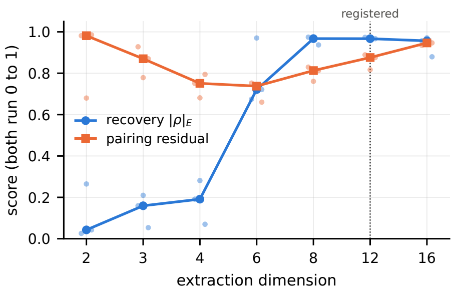

Figure 4: Energy recovery and pairing residual against extraction dimension. Lines are medians over the three models; points are individual models. Both quantities are dimensionless.

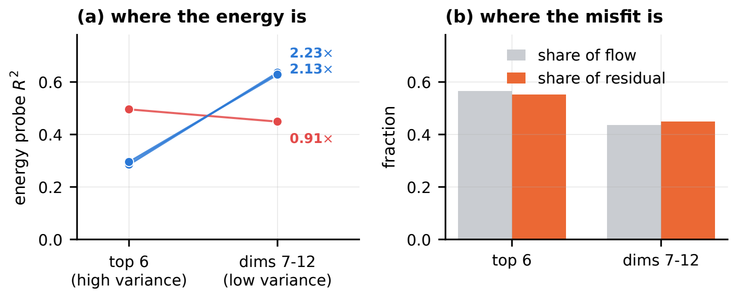

Figure 5: **(a)** Energy-probe R^2 for the top six principal directions against directions 7–12, per model. **(b)** Fraction of latent flow and pairing residual carried by each group, medians over three models.

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